

NC words

Following is a list of the different types of words in the formation of a block. Not every NC machine uses all the words. Also, the manner in which the words are expressed will differ between machines. By convention, the words in a block are given in the following order:

SEQUENCE NUMBER (n-words): This is used to identify the block.

PREPARATORY WORD (g-words): This word is used to prepare the controller for instructions that are to follow. For example, the word g02 is used to prepare the NC controller unit for circular interpolation along an arc in the clockwise direction. The preparatory word is needed so that the controller can correctly interpret the data that follow it in the block.

COORDINATES (x-, y-, and z-words): These give the coordinate positions of the tool. In a two-axis system, only two of the words would be used. In a four- or five-axis machine, additional a-words and/or b-words would specify the angular positions.

Although different NC systems use different formats for expressing a coordinate, we will adopt the convention of expressing it in the familiar decimal form: For example, x+7.235 or y-0.500. Some formats do not use the decimal point in writing the coordinate. The + sign to define a positive coordinate location is optional. The negative sign is, of course, mandatory.

FEED RATE (f-word): This specifies the feed in a machining operation. Units are inches per minute (ipm) by convention.

CUTTING SPEED (s-word): This specifies the cutting speed of the process, the rate at which the spindle rotates.

TOOL SELECTION (t-word): This word would be needed only for machines with a tool turret or automatic tool changer. The t-word specifies which tool is to be used in the operation. For example, t05 might be the designation of a 1/2-in. drill bit in turret position 5 on an NC turret drill.

MISCELLANEOUS FUNCTION (m-word): The m-word is used to specify certain miscellaneous or auxiliary functions which may be available on the machine tool. Of course, the machine must possess the function that is being called. An example would be m03 to start the spindle rotation. The miscellaneous function is the last



G & M Codes

Mill G Codes

G00	Rapid Positioning Motion (X,Y,Z,A,B)
G01	Linear Interpolation Motion (X,Y,Z,A,B,F)
G02	Circular Interpolation Motion CW (X,Y,Z,A,I,J,K,R,F)
G03	Circular Interpolation Motion CCW (X,Y,Z,A,I,J,K,R,F)
G04	Dwell (P) (P=Seconds". Milliseconds)
G09	Exact Stop, Non-Modal
G17*	Circular Motion XY Plane Selection (G02 or G03)
G18	Circular Motion ZX Plane Selection (G02 or G03)
G19	Circular Motion YZ Plane Selection (G02 or G03)
G20	Inch Coordinate Positioning
G21	Metric Coordinate Positioning
G28	Machine Zero Return Thru Ref. Point (X,Y,Z,A,B)
G29	Move to Location Through G28 Ref. Point (X,Y,Z,A,B)
G40*	Cutter Comp Cancel G41/G42/G141 (X,Y)
G41	2D Cutter Compensation, Left (X,Y,D)
G42	2D Cutter Compensation, Right (X,Y,D)
G43	Tool Length Compensation + (H,Z)
G49*	Tool Length Compensation Cancel G43/G44/G43
G52	Work Offset Positioning Coordinate
G53	Machine Positioning Coordinate, Non-Modal (X,Y,Z,A,B)
G54*	Work Offset Positioning Coordinate #1 (Setting 56)
G55	Work Offset Positioning Coordinate #2
G56	Work Offset Positioning Coordinate #3
G57	Work Offset Positioning Coordinate #4
G58	Work Offset Positioning Coordinate #5
G59	Work Offset Positioning Coordinate #6

G73	HS Peck Drilling Canned Cycle (X,Y,A,B,Z,I,J,K,Q,P,R,L,F)
G74	Reverse Tapping Canned Cycle (X,Y,A,B,Z,J,R,L,F)
G76	Fine Boring Canned Cycle (X,Y,A,B,Z,I,J,P,Q,R,L,F)
G77	Black Bore Canned Cycle (X,Y,A,B,Z,I,J,Q,R,L,F)
G80*	Cancel Canned Cycle (Setting 56)
G81	Drill Canned Cycle (X,Y,A,B,Z,R,L,F)
G82	Spot Drill / Counterbore Canned Cycle (X,Y,A,B,Z,P,R,L,F)
G83	Peck Drill Deep Hole Canned Cycle (X,Y,A,B,Z,I,J,K,Q,P,R,L,F)
G84	Tapping Canned Cycle (X,Y,A,B,Z,J,R,L,F)
G85	Bore In ~ Bore Out Canned Cycle (X,Y,A,B,Z,R,L,F)
G86	Bore In ~ Stop ~ Rapid Out Canned Cycle (X,Y,A,B,Z,R,L,F)
G87	Bore In ~ Manual Retract Canned Cycle (X,Y,A,B,Z,R,L,F)
G88	Bore In ~ Dwell ~ Manual Retract Canned Cycle (X,Y,A,B,Z,P,R,L,F)
G89	Bore In ~ Dwell ~ Bore Out Canned Cycle (X,Y,A,B,Z,P,R,L,F)
G90*	Absolute Positioning Command
G91	Incremental Positioning Command
G92	Global Work Coordinate System
G93	Inverse Time Feed Mode ON
G94	Inverse Time Feed OFF / Feed Per Minute ON
G98	Canned Cycle Initial Point Return
G99	Canned Cycle R Plane Return

Mill M Codes

M00	Program Stop
M01	Optional Program Stop
M02	Program End
M03	Spindle ON Clockwise (S)
M04	Spindle ON Counterclockwise (S))
M05	Spindle Stop
M06	Tool Change (T)
M08	Coolant ON
M09	Coolant OFF
M30	Program End and Reset
M31	Chip Auger Forward
M33	Chip Auger Stop
M34	Coolant Spigot Position Down, Increment
M35	Coolant Spigot Position Up, Decrement
M36	Pallet Part Ready
M41	Spindle Low Gear Override
M42	Spindle High Gear Override
M50	Execute Pallet Change
M83	Auto Air Jet ON
M84	Auto Air Jet OFF
M88	Coolant Through Spindle ON
M97	Local Sub-Program Call (P , L)
M98	Sub-Program Call (P , L)
M99	Sub-Program / Routine Return of Loop (P)

Notes:

[Cutting tool diameter = 20]

N01 G71 G90 G94

N02 G00 X-20 Y0

N03 G00 Z10

N04 G01 G42 D20 Z-10 S1000 F200 M03

N05 G01 X0 Y0

N06 G01 X90

N07 G01 Y90

N08 G01 X80 Y100

N09 G01 X10 Y

N10 G03 X0 Y90 R10

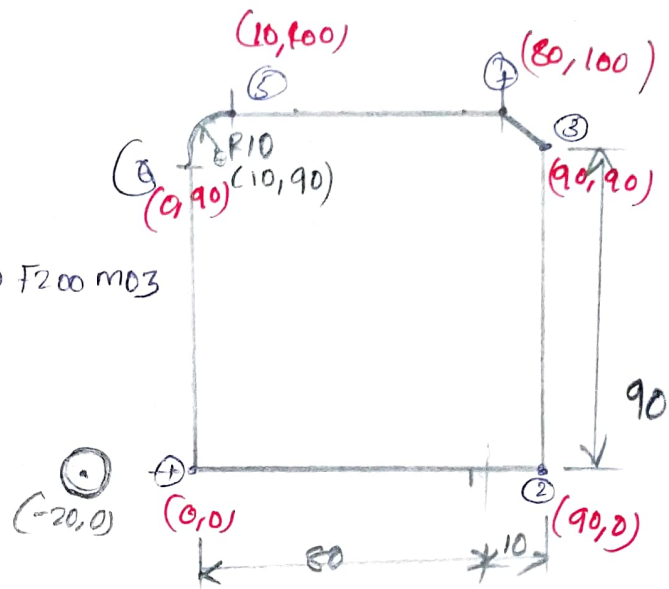
N11 G01 Y0

N12 G40

N13 G00 X-20 Y0

N14 G00 Z0

N15 M30



(X0 Y90 I0 J-10)

G28: Machine Zero (Home Point)

G81 - Drilling cycle

G80 - Drilling cycle cancel

G99 - Repeat Point return

G81 X-Y-Z-R-F

N01 G00 ^{Repeat Point} G28

N02 M06 T03 (Drill bit dia 10mm)

3 holes dia 10mm
through

N03 G90 G00 X10 Y10 Z10 S1100 M03 M08

N04 G99 G81 X10 Y10 Z-17 R2 F100

N05 X⁴⁰30 Y⁴⁰30

N06 X⁹⁰40 Y⁹⁰40

N07 G00 G80 Z10

N08 G00 M05 M09

N09 G00 G28

N10 M30

